

Лабораторная работа № 11

Настройка SSH-сервера

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Цель:

Настроить безопасный удалённый доступ, ключевую аутентификацию и SSH-туннели.



Успешный вход по паролю

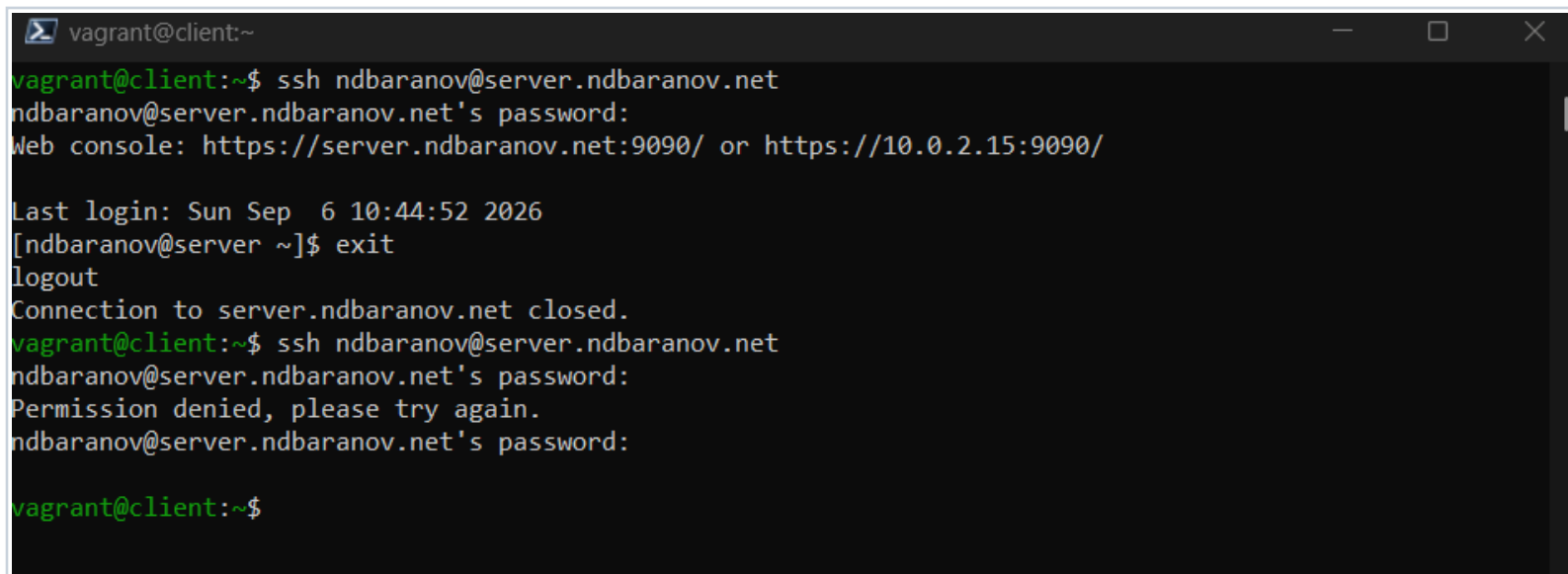
```
PS D:\work\ndbaranov\vagrant> vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 11:14:29 2026 from 10.0.2.2
vagrant@client:~$ ssh root@server.ndbaranov.net
The authenticity of host 'server.ndbaranov.net (192.168.1.1)' can't be established.
ED25519 key fingerprint is SHA256:iau18fCj0eSh49q6MHf6f97s45Fg260mmxn3sLjKgno.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'server.ndbaranov.net' (ED25519) to the list of known hosts.
root@server.ndbaranov.net's password:
Permission denied, please try again.
root@server.ndbaranov.net's password:
Permission denied, please try again.
root@server.ndbaranov.net's password:

vagrant@client:~$ ssh root@server.ndbaranov.net
root@server.ndbaranov.net's password:
Permission denied, please try again.
root@server.ndbaranov.net's password:
```

Смысл результата

С client выполнено SSH-подключение к server с корректными учётными данными. Приглашение оболочки подтверждает интерактивный вход.

Отклонение неверного пароля

A terminal window titled 'vagrant@client:~' with standard window controls. It shows a successful SSH login to 'ndbaranov@server.ndbaranov.net' followed by an 'exit' command. Then, a second SSH attempt is shown where the password is incorrect, resulting in a 'Permission denied' message.

```
vagrant@client:~$ ssh ndbaranov@server.ndbaranov.net
ndbaranov@server.ndbaranov.net's password:
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/

Last login: Sun Sep  6 10:44:52 2026
[ndbaranov@server ~]$ exit
logout
Connection to server.ndbaranov.net closed.
vagrant@client:~$ ssh ndbaranov@server.ndbaranov.net
ndbaranov@server.ndbaranov.net's password:
Permission denied, please try again.
ndbaranov@server.ndbaranov.net's password:

vagrant@client:~$
```

Смысл результата

Повторная попытка с ошибочным паролем завершается Permission denied. Это показывает, что сервер не предоставляет доступ без аутентификации.

Дополнительный порт sshd

```
ndbaranov@server:~  
vagrant@client:~$ ssh ndbaranov@server.ndbaranov.net  
ndbaranov@server.ndbaranov.net's password:  
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/  
  
Last login: Sun Sep  6 10:44:52 2026  
[ndbaranov@server ~]$ exit  
logout  
Connection to server.ndbaranov.net closed.  
vagrant@client:~$ ssh ndbaranov@server.ndbaranov.net  
ndbaranov@server.ndbaranov.net's password:  
Permission denied, please try again.  
ndbaranov@server.ndbaranov.net's password:  
  
vagrant@client:~$ ssh ndbaranov@server.ndbaranov.net  
ndbaranov@server.ndbaranov.net's password:  
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/  
  
Last failed login: Sun Sep  6 11:22:35 UTC 2026 from 192.168.1.31 on ssh:notty  
There was 1 failed login attempt since the last successful login.  
Last login: Sun Sep  6 11:20:04 2026 from 192.168.1.31  
[ndbaranov@server ~]$
```

Смысл результата

В sshd_config добавлен Port 2022 при сохранении стандартного порта 22. Проверка sshd -t подтверждает корректность файла.

Порт SSH в SELinux

```
root@server:~  
[root@server ~]# firewall-cmd --list-forward-ports  
port=2022:proto=tcp:toport=22:toaddr=  
[root@server ~]# firewall-cmd --remove-forward-port=port=2022:proto=tcp:toport=22  
firewall-cmd --permanent --remove-forward-port=port=2022:proto=tcp:toport=22  
firewall-cmd --reload  
success  
success  
success  
[root@server ~]# nano /etc/ssh/sshd_config  
[root@server ~]# sshd -t  
[root@server ~]# systemctl restart sshd  
[root@server ~]# systemctl status -l sshd --no-pager  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)  
   Active: active (running) since Sun 2026-09-06 11:24:56 UTC; 5s ago  
  Invocation: 38ff3621ecda485cb4d0ef2efdbba463  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
   Main PID: 20632 (sshd)  
      Tasks: 1 (limit: 10232)  
    Memory: 1.3M (peak: 1.5M)  
       CPU: 17ms  
    CGroup: /system.slice/sshd.service  
            └─20632 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Sep 06 11:24:56 server systemd[1]: Starting sshd.service - OpenSSH server daemon...  
Sep 06 11:24:56 server sshd[20632]: error: Bind to port 2022 on 0.0.0.0 failed: Permission denied.  
Sep 06 11:24:56 server sshd[20632]: error: Bind to port 2022 on :: failed: Permission denied.  
Sep 06 11:24:56 server sshd[20632]: Server listening on 0.0.0.0 port 22.  
Sep 06 11:24:56 server systemd[1]: Started sshd.service - OpenSSH server daemon.  
Sep 06 11:24:56 server sshd[20632]: Server listening on :: port 22.  
[root@server ~]#
```

Смысл результата

Командой semanage порт 2022 связан с типом ssh_port_t. Без этого SELinux мог бы запретить sshd прослушивать нестандартный порт.

Правило firewalld

```
Выбрать root@server:~
root@server ~]# command -v semanage
/sbin/semanage
root@server ~]# semanage port -a -t ssh_port_t -p tcp 2022
root@server ~]# semanage port -l | grep ssh_port_t
ssh_port_t                tcp                2022, 22
root@server ~]# firewall-cmd --add-port=2022/tcp
firewall-cmd --add-port=2022/tcp --permanent
success
success
root@server ~]# systemctl restart sshd
root@server ~]# ss -ltnp | grep sshd
LISTEN 0      128          0.0.0.0:2022  0.0.0.0:*    users:((("sshd",pid=20680,fd=7))
LISTEN 0      128          0.0.0.0:22   0.0.0.0:*    users:((("sshd",pid=20680,fd=9))
LISTEN 0      128          [::]:2022   [::]:*       users:((("sshd",pid=20680,fd=8))
LISTEN 0      128          [::]:22     [::]:*       users:((("sshd",pid=20680,fd=10))
root@server ~]#
```

Смысл результата

TCP-порт 2022 разрешён в межсетевом экране. Команда `firewall-cmd` выводит новое постоянное правило.

Прослушивание портов

```
ndbaranov@server:~  
[ndbaranov@server ~]$ ssh ndbaranov@server.ndbaranov.net  
The authenticity of host 'server.ndbaranov.net (192.168.1.1)' can't be established.  
ED25519 key fingerprint is SHA256:iau18fCj0eSh49q6MHf6f97s45Fg260mmxn3sLjKgn0.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'server.ndbaranov.net' (ED25519) to the list of known hosts.  
ndbaranov@server.ndbaranov.net's password:  
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/  
  
Last login: Sun Sep  6 11:23:27 2026 from 192.168.1.31  
[ndbaranov@server ~]$ exit  
logout  
Connection to server.ndbaranov.net closed.  
[ndbaranov@server ~]$ ssh -p 2022 ndbaranov@server.ndbaranov.net  
ndbaranov@server.ndbaranov.net's password:  
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/  
  
Last login: Sun Sep  6 11:26:34 2026 from 192.168.1.1  
[ndbaranov@server ~]$ sudo -i  
[sudo] password for ndbaranov:  
[root@server ~]# exit  
logout  
[ndbaranov@server ~]$ exit  
logout  
Connection to server.ndbaranov.net closed.  
[ndbaranov@server ~]$
```

Смысл результата

Утилита ss показывает sshd на портах 22 и 2022. Таким образом подтверждена работа не только конфигурации, но и сетевого сокета.

Создание RSA-ключа

```
ndbaranov@server:~  
Last login: Sun Sep  6 10:39:38 UTC 2026 on pts/0  
ndbaranov@client:~$ ssh-keygen -t rsa  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/ndbaranov/.ssh/id_rsa):  
Created directory '/home/ndbaranov/.ssh'.  
Enter passphrase for "/home/ndbaranov/.ssh/id_rsa" (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /home/ndbaranov/.ssh/id_rsa  
Your public key has been saved in /home/ndbaranov/.ssh/id_rsa.pub  
The key fingerprint is:  
SHA256:gxndBo6lTvypbtPOXGugcikYvNia4IBfnBDitPS/NU8 ndbaranov@client.ndbaranov.net  
The key's randomart image is:  
+---[RSA 3072]-----+  
|.o.o  
|+oo* o  
| o*.. =  
|.o.+. =  
|= . *.ooSE  
|=o.o.o +.  
|o+++ o. .  
|=+Bo...  
|o*.o+..  
+----[SHA256]-----+  
ndbaranov@client:~$ ~/.ssh/id_rsa  
~/.ssh/id_rsa.pub  
-bash: /home/ndbaranov/.ssh/id_rsa: Permission denied  
-bash: /home/ndbaranov/.ssh/id_rsa.pub: Permission denied  
ndbaranov@client:~$ ls -l ~/.ssh/  
total 8  
-rw----- 1 ndbaranov ndbaranov 2622 Sep  6 11:30 id_rsa  
-rw-r--r-- 1 ndbaranov ndbaranov 584 Sep  6 11:30 id_rsa.pub  
ndbaranov@client:~$ ssh-copy-id ndbaranov@server.ndbaranov.net  
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/ndbaranov/.ssh/id_rsa.pub"  
The authenticity of host 'server.ndbaranov.net (192.168.1.1)' can't be established.  
ED25519 key fingerprint is SHA256:iaul8fCj0eSh49q6Mhf6f97s45Fg260mmxn3sLjKgn0.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already  
installed  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install  
the new keys  
ndbaranov@server.ndbaranov.net's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with: "ssh 'ndbaranov@server.ndbaranov.net'"  
and check to make sure that only the key(s) you wanted were added.  
  
ndbaranov@client:~$ ssh ndbaranov@server.ndbaranov.net  
Web console: https://server.ndbaranov.net:9090/ or https://10.0.2.15:9090/  
  
Last failed login: Sun Sep  6 11:29:16 UTC 2026 from 192.168.1.1 on ssh:notty  
There was 1 failed login attempt since the last successful login.  
Last login: Sun Sep  6 11:26:50 2026 from 192.168.1.1  
[ndbaranov@server ~]$
```

Смысл результата

На клиенте создана пара RSA-ключей.
Закрытый ключ остаётся у пользователя,
а открытая часть предназначена для
сервера.

Копирование открытого ключа

```
ndbaranov@client:~  
ndbaranov@client:~$ lsof | grep TCP  
lsof: WARNING: can't stat() fuse.portal file system /run/user/42/doc  
Output information may be incomplete.  
ndbaranov@client:~$ ssh -fNL 8080:localhost:80 ndbaranov@server.ndbaranov.net  
ndbaranov@client:~$ lsof -iTCP:8080 -sTCP:LISTEN  
COMMAND  PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME  
ssh      26554 ndbaranov  4u  IPv6  83872    0t0  TCP localhost:webcache (LISTEN)  
ssh      26554 ndbaranov  5u  IPv4  83873    0t0  TCP localhost:webcache (LISTEN)  
ndbaranov@client:~$
```

Смысл результата

ssh-copy-id добавил ключ в authorized_keys пользователя на server. Права каталога .ssh сохранены безопасными.

Вход без пароля

```
Выбрать ndbaranov@client:~
ndbaranov@client:~$ lsof | grep TCP
lsof: WARNING: can't stat() fuse.portal file system /run/user/42/doc
Output information may be incomplete.
ndbaranov@client:~$ ssh -fNL 8080:localhost:80 ndbaranov@server.ndbaranov.net
ndbaranov@client:~$ lsof -iTCP:8080 -sTCP:LISTEN
COMMAND  PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
ssh      26554 ndbaranov  4u  IPv6  83872    0t0  TCP localhost:webcache (LISTEN)
ssh      26554 ndbaranov  5u  IPv4  83873    0t0  TCP localhost:webcache (LISTEN)
ndbaranov@client:~$ curl http://localhost:8080
Welcome to the server.ndbaranov.net server.
ndbaranov@client:~$
```

Смысл результата

Повторное подключение выполняется по открытому ключу без запроса пароля учётной записи. Сервер подтвердил владение соответствующим закрытым ключом.

Локальный SSH-туннель

```
ndbaranov@client:~$ ssh ndbaranov@server.ndbaranov.net hostname
server
ndbaranov@client:~$ ssh ndbaranov@server.ndbaranov.net ls -Al
total 20
-rw-----. 1 ndbaranov ndbaranov 569 Sep  6 11:30 .bash_history
-rw-r--r--. 1 ndbaranov ndbaranov 18 Oct 29 2024 .bash_logout
-rw-r--r--. 1 ndbaranov ndbaranov 144 Oct 29 2024 .bash_profile
-rw-r--r--. 1 ndbaranov ndbaranov 549 Sep  5 18:15 .bashrc
drwx-----. 2 ndbaranov ndbaranov  6 Sep  5 18:24 .cache
drwx-----. 5 ndbaranov ndbaranov 4096 Sep  6 10:44 Maildir
drwxr-xr-x. 4 ndbaranov ndbaranov  39 Sep  3 07:10 .mozilla
drwx-----. 2 ndbaranov ndbaranov 103 Sep  6 11:29 .ssh
ndbaranov@client:~$ ssh ndbaranov@server.ndbaranov.net 'MAIL=~/.Maildir/ mail'
s-nail version v14.9.24. Type '?' for help
/home/ndbaranov/Maildir: 3 messages 2 unread
  1 Super User      2026-09-06 10:02 17/664 "IMAP test 1"
  U 2 Super User    2026-09-06 10:02 17/666 "IMAP test 3"
  U 3 ndbaranov@client.ndb 2026-09-06 10:44 21/811 "LMTP test"
quit
Held 3 messages in /home/ndbaranov/Maildir
ndbaranov@client:~$
```

Смысл результата

Создано перенаправление
8080:localhost:80 через SSH. Локальный
порт клиента связан с HTTP-службой
сервера внутри зашифрованного канала.

Проверка туннеля и provision

```
Выбрать vagrant@server:~
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server ssh"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server ssh (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell20260906-22272-gh910c.sh
server: Provisioning SSH server
server: Install needed packages
server: Last metadata expiration check: 2:45:58 ago on Sun 06 Sep 2026 08:51:38 AM UTC.
server: Package policycoreutils-python-utils-3.10-2.el10_2.noarch is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: Configure sshd
server: Configure SELinux
server: Configure firewall
server: Warning: ALREADY_ENABLED: 2022:tcp
server: success
server: success
server: Check SSH configuration
server: Restart sshd
server: SSH provisioning completed
PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 11:15:31 2026 from 10.0.2.2
vagrant@server:~$ sudo ss -ltnp | grep sshd
LISTEN 0      128          0.0.0.0:2022  0.0.0.0:*    users:(("sshd",pid=21783,fd=7))
LISTEN 0      128          0.0.0.0:22   0.0.0.0:*    users:(("sshd",pid=21783,fd=9))
LISTEN 0      128          [::]:2022   [::]:*       users:(("sshd",pid=21783,fd=8))
LISTEN 0      128          [::]:22     [::]:*       users:(("sshd",pid=21783,fd=10))
vagrant@server:~$
```

Смысл результата

Запрос к localhost:8080 возвращает страницу Apache, после чего provision-сценарий закрепляет SSH-настройки. Туннель доказал практическую работу перенаправления.

Результат

SSH принимает подключения на стандартном и дополнительном портах, поддерживает вход по ключу и локальное перенаправление портов к веб-службе.